RoboCode Competition – Robot Design & Strategy Log

Team Name: SetuCW

Robot Name: Potujniy

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1. Brainstorming

Key Questions Considered:

- Type of robot: Hybrid.

- Strengths: Good moving and accurate targeting.

2. Robot Overview

2.1 High-Level Strategy

The robot patrols around the center of the battlefield using a square path. While no enemy is detected, it keeps scanning the center area. Once an enemy is found, it calculates the enemy's position, rotates the gun toward it, and fires.

2.2 Core Behaviours

Behaviour	**Description**	**Status (Planned / Implemented / Modified)**
Movement	Square patrol around center	Implemented
Targeting	Calculates enemy coordinates	Implemented
Firing	Distance-based firepower	Implemented
Evasion	-	-
Collision Handling	-	-

3. Technical Design

3.1 Key Algorithms & Logic

3.1.1 Movement Logic

The robot moves between four coordinate points. If an enemy is detected, rotates the gun toward it, and fires.

This allows the robot to control space efficiently and stay in a strong tactical position.

3.1.2 Targeting Logic

When the radar scans an opponent, the robot calculates its coordinates using angle and distance. The gun is then rotated precisely toward that position.

Direct coordinate calculation gives better accuracy than simple angle-based shooting.

3.1.3 Firing Logic

The robot fires only when the gun has cooled down.

Firepower is calculated based on distance. Closer targets receive stronger shots, while distant targets receive weaker ones. Power is always kept within safe limits.

4. Change Log & Iterative Development

Change Made:

Added coordinate-based targeting.

Expected Impact:

Higher hit rate.

Actual Outcome:

Shots became noticeably more consistent.

Lessons Learned:

Math-based targeting is more reliable than simple tracking.